

- | Modulation Scheme | Probability of error |
|----------------------|--|
| I. Coherent PSK | $A. \frac{1}{2} \operatorname{erfc}\left(\sqrt{\frac{E_b}{2N_o}}\right)$ |
| II. Coherent FSK | $B. \frac{1}{2} \operatorname{erfc}\left(\sqrt{\frac{E_b}{N_o}}\right)$ |
| III. DPSK | $C. \frac{1}{2} \exp\left(-\frac{E_b}{2N_o}\right)$ |
| IV. Non coherent FSK | $D. \frac{1}{2} \exp\left(-\frac{E_b}{N_o}\right)$ |

- A) I and II only
B) I and III only
C) II and III only
D) I, II and III
- The properties used for pseudorandom sequence are**
- A) Balance
B) Run
C) Correlation
D) All the mentioned

18. In direct sequence process which step is performed first?

- A) De-spreading
B) Demodulation
C) De-spreading & Demodulation
D) None of the mentioned

19. Which system allows larger processing gain?

- A) Direct sequence
B) Frequency hopping
C) Direct sequence & Frequency hopping
D) None of the mentioned

20. Advantages of using Spread Spectrum modulation is/are:

1. Interference rejection capability
2. Frequency planning is not required
3. Resistance to multipath fading
4. ISI is lesser

- A) 1 and 2 are correct
B) 1, 2 and 3 are correct
C) 2 and 3 are correct
D) All the four are correct

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

Marks BT* CO* PO*

1. a) For the finite energy signal as shown in Figure Q1(a), (i) sketch the impulse response $h_{opt}(t)$ of the optimum filter matched to $\phi(t)$ (ii) Determine the value of the output filter at $t = T_s$ assuming noise is zero and input is $\phi(t)$.

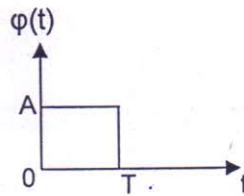


Figure Q1(a)

- b) Discuss the Nyquist's criterion for distortionless baseband binary transmission (zero ISI).
2. a) With neat diagrams explain the principle of detection and estimation in a typical digital communication system.
- b) For the binary sequence 011010111011 plot the following waveform formats:

- (i) NRZ polar
- (ii) RZ Bipolar
- (iii) Differential encoding with $d_1 = 1$
- (iv) NRZ Unipolar

3. a) Explain how a raised cosine spectrum can be used to reduce ISI.
- b) Consider the following binary signals defined in terms of orthonormal basis functions $\phi_1(t)$ and $\phi_2(t)$:
- (i) $s_1(t) = A \phi_1(t)$, $s_2(t) = A \phi_2(t)$
 - (ii) $s_1(t) = A \phi_1(t)$, $s_2(t) = -A \phi_2(t)$
- Draw the constellation diagram, maximum likelihood decision regions, write decision rules for the given signal sets and also draw the diagram for the optimum receiver.

Unit – II

4. a) Derive the expression for error probability for the coherent BFSK scheme.

8	L*3	1	1
8	L2	2	1
8	L2	1	1
8	L3	2	1
8	L2	2	1
8	L3	1	1
8	L2	4	1

- b) For the coherent binary PSK scheme: (i) determine the signal points (ii) basis functions (iii) draw the constellation diagram and mark decision regions (iv) draw the transmitter and receiver structure. 8 L2 3 1
5. a) Explain the ASK scheme with necessary equations, constellation diagram and transmitter, receiver structure. 8 L2 4 1
- b) Explain M-ary QAM scheme and for $M = 16$ draw the constellation diagram. 8 L2 3 1
6. a) Explain the following:
(i) QPSK (ii) DPSK 10 L2 4 1
- b) Binary data is transmitted over AWGN channel using BPSK at a rate of 1 Mbps. It is desired to have average probability of error $P_e \leq 10^{-4}$. Noise PSD is $N_0/2 = 10^{-12}$ watt/Hz. Determine the average carrier power required at receiver input if the detector is of coherent type. Assume $\text{erfc}(3.5) = 0.00025$. 6 L3 3 1
- Unit – III**
7. a) Explain the frequency hopped spread spectrum scheme with neat diagrams and discuss the processing gain. 8 L2 5 1
- b) Explain the role of PN sequence in spread spectrum techniques. List the properties of maximal length sequence. A 4-stage LFSR generates the sequence 111101011001000111101.... Verify any two properties and determine the period of the PN sequence. 8 L3 5 1
8. a) Explain the direct sequence spread spectrum scheme with neat diagrams and necessary equations. 8 L2 5 1
- b) Consider the three-stage shift register as shown in Figure Q8(b) below. Assume the initial state of the shift register as 100. Determine the PN sequence generated and examine all the properties.

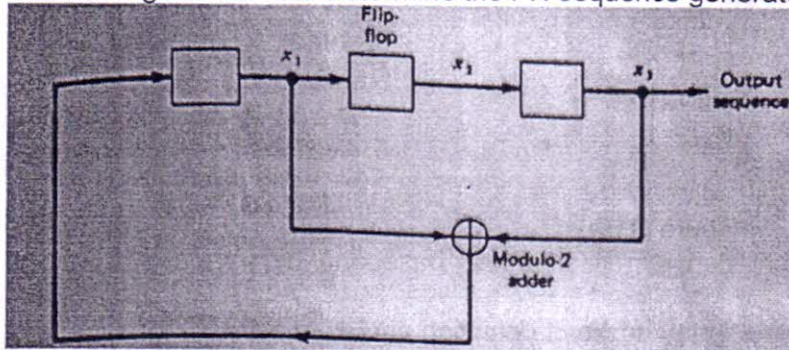


Figure Q8(b)

8 L3 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
